PHASE 6 DOCUMENT

EduGenie: Google Gemini Powered Learning Assistant

Phase Name: Project Assistance, Faculty Support and Resource Management

Project Name	EduGenie: Google Gemini Powered Learning Assistant
Phase	Phase 6 – Project Assistance and Faculty Support
Project Type	AI-Powered Learning Application
Main Technology	Google Gemini API
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 6 of **EduGenie: Google Gemini Powered Learning Assistant**. This phase expands the platform beyond individual learning support by adding project-assistance tools, faculty-supported learning resources, and controlled academic guidance. The phase enables students to use EduGenie for structured project planning while allowing faculty members to share approved resources and learning prompts.

2. Phase Objective

The objective of this phase is to help students plan and document academic projects responsibly, while preserving faculty oversight. EduGenie will support students with project titles, problem statements, objectives, module ideas, documentation outlines, technology suggestions, and presentation preparation. Faculty users can publish approved learning resources, suggested prompts, and project guidelines for their students.

3. Project Assistance Module

The Project Assistance module provides guided AI support for academic projects. Students can choose a project domain, enter a topic or problem area, and request ideas such as project titles, abstracts, objectives, scope, proposed modules, system architecture suggestions, development plans, testing ideas, and future enhancements. The purpose is to help students organize their work, not to generate a complete submission without student understanding and faculty review.

4. Documentation Support

EduGenie can help students prepare structured documentation for a project. A student may request an abstract, introduction, problem statement, literature-review outline, requirement list, module description, methodology, testing plan, conclusion, or presentation outline. The tool should encourage students to edit the generated draft, add their own project-specific evidence, and verify that the final document reflects actual work completed by the team.

5. Project Planning Workflow

1. Student opens the Project Assistance feature.
2. Student selects or enters the project domain and topic.
3. Student chooses the required support type, such as title ideas, objectives, module plan, documentation outline, or presentation help.
4. EduGenie sends a structured request to the secure Gemini backend.
5. The system displays the generated guidance.
6. Student reviews, edits, saves, or asks a follow-up question.
7. Student discusses important decisions with the faculty guide.
8. The team uses the approved plan to develop the actual project.

6. Faculty Resource Management

The Faculty Resource Management module allows authorized faculty members to share approved academic material with students. Resources can include course notes, project guidelines, reference links, recommended reading, assessment instructions, approved prompt templates, or model documentation structures. Faculty resources can be organized by subject, department, class, topic, or project category.

7. Faculty Guidance Features

Faculty members may publish learning announcements, share topic-specific prompts, recommend revision activities, and provide project instructions. They can guide students toward approved resources rather than depending only on open-ended AI responses. Where enabled, faculty may view high-level, privacy-respecting indicators such as common topics requested by a class or the number of students using a resource, without accessing unnecessary private conversation content.

8. Student Access to Resources

Students can access only the resources relevant to their subject, class, department, or assigned project category. The resource screen should show a title, description, resource type, faculty name where appropriate, and access link or attachment. Students can view, save, or mark a resource as useful. Access control should prevent students from editing faculty material or viewing resources outside their authorized scope.

9. Project Guidance Boundaries

EduGenie should assist students in understanding and planning work, but it should not replace the academic contribution of the student team. The system should discourage plagiarism, direct copying, fabricated results, and false claims about implementation. It should encourage students to cite sources where required, test their own system, write documentation based on their actual development, and obtain faculty approval for project decisions.

10. Resource and Project Data Design

This phase introduces ProjectIdeas, ProjectDrafts, DocumentationTemplates, FacultyResources, ResourceCategories, Announcements, and ResourceAccessLogs. Project-related drafts should be associated with the correct student or team account. Faculty resources should include ownership, target audience, visibility status, publication date, and access permissions. The database should support editing and version tracking where required.

11. User Interface Design

The student Project Assistance screen can include cards for Project Title, Abstract, Objectives, Modules, Documentation, Testing, and Presentation. The Faculty Resources screen can provide filters for subject, type, and category. Faculty users need a clear interface to create, upload, publish, edit, archive, or remove resources according to their permissions. The design should show whether an item is AI-generated guidance, faculty-approved material, or student-saved content.

12. Security and Content Control

Only authorized faculty users should be able to publish or change shared resources. The system should validate attachments and external links where applicable, prevent unauthorized editing, and log important publishing actions. Student project drafts should be private by default unless shared intentionally with a team or faculty guide. The application should protect user data and avoid exposing one student team’s work to another team.

13. Testing Plan

Testing should confirm that students can request project guidance, receive relevant structured responses, save their own drafts, and access only authorized resources. Faculty testing should verify resource creation, publication, editing, visibility rules, and removal. The system should be tested for unauthorized access, invalid links, empty inputs, AI-service failures, mobile responsiveness, and clear distinction between faculty resources and AI-generated guidance.

14. Phase Deliverables

The deliverables of this phase are the Project Assistance module, structured Gemini prompt templates for project guidance, documentation-support features, student project-draft storage, Faculty Resource Management interface, resource publishing and access-control rules, faculty guidance features, security controls, and test evidence for student and faculty workflows.

15. Planned Next Phase

The next phase will be **Phase 7: System Testing, Quality Assurance and Security Validation**. It will test all completed EduGenie modules end-to-end, document test cases and results, verify role-based access and Gemini integration behavior, identify defects, and prepare the application for final deployment and project evaluation.

16. Conclusion

Phase 6 adds practical academic value to EduGenie by supporting student project planning and connecting AI assistance with faculty-approved resources. The phase promotes structured project development, responsible use of AI, and meaningful faculty involvement. With these features in place, EduGenie is ready for comprehensive testing and quality validation.

Phase Approval

Prepared by	_____
Guide / Faculty	_____
Approved by	_____